TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Phạm Minh Trí		tripm14@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Trương Hoàng Quí	SE184355	0862358874	QuiTHSE184355@fpt.edu.vn	Leader
2	Khúc Ngọc Sơn	SE184040	0981761545	SonKNSE184040@fpt.edu.vn	Member
3	Cao Hữu Trí	SE184047	0356702429	TriCHSE184047@fpt.edu.vn	Member
4	Lê Thị Trà Mi	SE184379	0388278354	MiLTTSE184379@fpt.edu.vn	Member
5	Nguyễn Văn Sang	SE183276	0983135379	SangNVSE183276@fpt.edu.vn	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***HR Connect: AI-Powered Recruitment & Affiliate Headhunting Platform***

3.1.2. Vietnamese: ***HR Connect: Nền tảng Tuyển dụng và Mạng lưới Affiliate Headhunter tích hợp AI***

(*) 3.2. Main proposal content (including result and product)

a) Context:

Small and medium recruitment agencies in Vietnam, typically operate three service lines at once: commission-based headhunting (COD - Cash/Commission on Delivery), CV sourcing for clients without in-house recruiters, and open CV-application services for clients who want to receive candidates directly. In practice these three workflows are still coordinated manually through spreadsheets, Zalo groups, and email, which creates duplicated candidate submissions, unclear ownership of a candidate between competing recruiters, and no reliable way to track a

candidate through screening, interview, ~~offer, and probation-period warranty~~offer, start of work and the post-placement warranty.

A second structural problem is that this business model depends on a distributed network of freelance recruiters and referral partners - internally called OPR Hub (affiliate headhunters) - who submit candidates for open jobs in exchange for commission ~~once a candidate passes probation~~once a placed candidate completes the warranty period and the client has paid. Without a shared system, the agency cannot verify who submitted a candidate first (leading to commission disputes), cannot rate the historical quality of each affiliate's submissions, and cannot give affiliates real-time visibility into which jobs are open, urgent, or already filled.

A third gap is on the matching side: with dozens of jobs and hundreds of CVs arriving weekly, HR staff manually re-read every resume against the job description, which is slow and inconsistent across reviewers. There is a clear opportunity to apply AI-assisted candidate-job matching to pre-screen and rank submitted CVs, while keeping a human reviewer in the loop for the final decision - directly addressing a real operational bottleneck rather than a hypothetical one.

b) Proposed Solutions:

HR Connect is proposed as a role-based web platform that unifies three actor groups - client companies, candidates, and affiliate/internal recruiters - around a single job and candidate pipeline, with an AI matching engine that scores every submitted CV against its target job description and an affiliate/commission ledger that removes ambiguity from who introduced which candidate.

Workflow:

Client registers company profile & posts a job -> ~~Admin approves & publishes job~~Internal HR reviews the job and agrees the commercial terms -> Client signs the service agreement by OTP (Headhunt / Sourcing) -> posting fee or deposit recorded -> job published -> Affiliate/HH/Candidate submits CV to job -> ~~System deduplicates by email/phone~~Candidate consents to an affiliate referral by e-mail -> System deduplicates by email/phone -> AI Matching Engine scores CV against JD -> Admin/Client reviews color-coded ranked shortlist -> Interview scheduling & outcome logging -> ~~Offer (OB) & Probation (PP) tracking -> Commission calculated & paid to affiliate~~Offer (OB) & start of work -> Warranty tracking -> Commission earned after the warranty and the client's payment, then paid to the affiliate -> Feedback rating recorded for affiliate reliability

The system is designed around HR Connect existing three service lines (Headhunt COD, CV Sourcing, CV Application) so that each job can be tagged with the service type the client selected, and the same candidate pipeline serves headhunters, affiliates, and self-service candidate applications without divergent code paths.

Human Actors

- ~~System Administrator (Admin/HR) - approves company and job postings, reviews and forwards candidates, resolves duplicate-submission disputes, tracks interview outcomes and warranty periods, and manages affiliate commission payouts.~~Internal HR - reviews and approves job postings and their commercial terms, sends service agreements, screens candidates for Headhunt and CV Sourcing jobs, delivers Sourcing CVs and verifies resignations during the warranty. Internal HR does not submit CVs on behalf of candidates.

● Platform Admin - approves companies and affiliates, manages users, roles and permissions, service types, commission rules and business settings, records fees, deposits and payouts, and resolves source disputes.

● Client Company - registers a company profile, selects a service package (Headhunt / Sourcing / Application), posts and manages job requisitions, reviews the AI-ranked candidate shortlist, and records interview feedback.

● Candidate - creates a profile/CV via three options (platform builder, template form, or file upload), browses and applies to job postings, and receives AI-generated job-fit notifications.

● Affiliate Recruiter / OPR Hub Freelancer - browses open/active jobs relevant to their specialty, submits candidates against specific jobs to earn commission, and tracks their submission-to-hire performance and payout history on a personal dashboard.

Five Core Modules

● 1. Company & Job Management - company profile registration/approval, job posting creation (Details, Instructions, Objectives, Engagement sections), ~~job status lifecycle (Draft/Active/Closed), and public/affiliate-only job visibility rules.~~job status lifecycle (Draft, Pending Review, Pending Agreement, Pending Payment, Active, Paused, Closed, Rejected), OTP-signed service agreements for Headhunt and CV Sourcing, and job visibility and submission rights configured per role and service type.

● 2. Candidate & CV Management - candidate account and profile creation (build-on-platform, template-based, or upload), ~~automatic duplicate detection by email/phone,~~automatic duplicate detection by email/phone, candidate consent by e-mail for every affiliate referral, and a Candidate Highlight summary (current salary, expected salary, experience, language proficiency, availability date).

● 3. AI Matching & Scoring Engine - ~~NLP-based similarity scoring between a candidate's CV/profile and a job description,~~Sentence-BERT similarity plus an LLM assessment of every must-have and should-have requirement with evidence quoted from the CV and verified, computed asynchronously, producing a color-coded match tier (>=80% red, 70-79% teal, 60-69% light green, <60% gray) to prioritize human review.

● 4. Affiliate & Commission Management (OPR Hub) - affiliate registration, job-to-candidate attribution and first-submission timestamping to prevent disputes, ~~commission-rule configuration (probation-pass, contract-signed milestones),~~source disputes resolved by the Platform Admin, commission-rule configuration (warranty-passed milestone for Headhunt, accepted-CV milestone for CV Sourcing), payout ledger, and affiliate quality rating based on submission-to-hire conversion.

● 5. Workflow, Interview & Reporting - interview scheduling and outcome notes (Pass/Fail/Backup), ~~offer (OB) and probation (PP) status tracking with warranty-period countdown,~~offer (OB) and start-of-work tracking with warranty-period countdown, resignation reports and free replacement slots, and a reporting dashboard (jobs by status, top affiliates, top companies, revenue by service line).

The platform is designed as a responsive web application with a public marketing site (jobs, company directory, service pages) and authenticated dashboards per role, backed by a relational database for transactional data and a microservice for the AI scoring engine.

c) Functional Requirements:

Job Management

- Create job/JD with description, requirements, and benefits
- ~~Configure Must-have (mandatory) and Should-have (preferred) keyword sets for automatic pre-filtering~~Define Must-have (mandatory) and Should-have (preferred) requirements that the AI engine assesses one by one
- ~~Approve and publish jobs~~Review jobs and agree commercial terms, then publish them to make them visible on the public site and affiliate portal
- Send a versioned service agreement for Headhunt and CV Sourcing jobs; the client accepts or requests changes and signs it with an e-mail OTP
- ~~Track job status: Active / Paused / Closed / Draft,~~Track job status: Draft / Pending Review / Pending Agreement / Pending Payment / Active / Paused / Closed / Rejected, with 1-job-open-at-a-time and 2-month reopen cycle rules for free-tier companies

Application & CV Management

- Submit CV to a specific job (self-service candidate, affiliate, or freelance headhunter)
- Automatic duplicate check by email/phone to prevent multiple submissions of the same candidate
- Ask the candidate to consent by e-mail before an affiliate referral enters the pipeline; let affiliates dispute a duplicate or attribution result
- Preliminary CV screening against job requirements/notes/skills
- Approve or reject a submitted CV, with a mandatory rejection reason logged for audit

AI Matching Engine

- Score every submitted CV against its target JD and return a percentage match
- Assign a color-coded action tier so reviewers can triage at a glance (>=80% red heart, 70-79% teal, 60-69% light green, <60% gray)
- Auto-generate a Candidate Highlight card: current salary, expected salary, years of experience with a headline achievement, language level, and availability date
- Route headhunt-service jobs to HR staff for candidate contact, and application/sourcing-service jobs to the client's own HR for contact

Workflow & Interview

- Coordinate interview scheduling and send notifications to candidate and interviewer
- Record post-interview outcome notes (Pass / Fail / Backup)
- ~~Register offer acceptance (OB) and probation start (PP), and count down the agreed warranty/probation period~~Register offer acceptance (OB) and the actual start of work, count down the agreed warranty period, and handle resignations with a free replacement slot

Partner & Finance (Affiliate / OPR Hub)

- Maintain a directory of affiliate/freelance recruiters with their submission history and performance stats
- Attribute each hired candidate to the affiliate who first submitted them, timestamped to resolve disputes
- ~~Automatically calculate commission at defined milestones (probation passed, contract signed)~~Automatically calculate commission at defined milestones (warranty passed for Headhunt, CV accepted for CV Sourcing), payable after the client's payment
- Track the candidate warranty period per placement and its effect on final commission release

● Maintain an affiliate quality rating (feedback/rating) based on submission-to-hire conversion rather than raw volume

Reporting & Company Directory

● Publish a searchable/filterable company directory (by industry, using the LinkedIn L1/L2/L3 industry taxonomy) with an admin-curated Top Companies list

● Filter job listings by region, salary range, years of experience, and industry

● Dashboard summaries: jobs by status, top-performing affiliates by quarter/year, revenue by service line (Headhunt / Sourcing / Application)

d) Non-Functional Requirements:

Performance

● Job search and filter results return within fast time for a candidate pool of at least 50,000 profiles

● ~~AI matching score for a single CV/JD pair computed within 5 seconds~~AI matching runs asynchronously; 95% of applications receive their score within 60 seconds of submission

Security

● Role-based access control separating Admin, Client, Candidate, and Affiliate views/data

● Encrypted storage of personal candidate data (CV files, salary, contact info) in compliance with Vietnam's Personal Data Protection Decree

Reliability

● Duplicate-submission check must run synchronously before a CV is accepted into the pipeline, with accuracy on exact email/phone matches

Scalability

● Architecture must support horizontal scaling of the AI scoring microservice independently of the core transactional backend during peak posting periods

Maintainability

● Configurable keyword sets, commission rules, and industry taxonomy (backed by the 785-row Industry/Company-Size reference table) without code changes

Transparency & Auditability

● Every candidate-to-affiliate attribution and commission calculation must be logged with an immutable timestamp and reviewable by Admin

e) Theory & Practical:

Applied Theory

● Software Engineering: full SDLC applied to a multi-role marketplace platform, including use-case modeling, ERD design for a transactional recruitment schema, and Agile/Scrum sprint planning across a ~~4-person team~~5-person team.

● Object-Oriented Analysis & Design: domain modeling of Job, Candidate, Application, Affiliate, and Commission entities and their state-machine lifecycles (Job: Draft->Active->Closed; Application: Submitted->Screened->Interviewed->Offered->Hired).

● Information Retrieval & NLP: TF-IDF/embedding-based semantic similarity (e.g., Sentence-BERT) for CV-to-JD matching, ~~and keyword-based Must-have/Should-have rule filtering as a complementary, explainable pre-filter.~~combined with an LLM that assesses each Must-

have/Should-have requirement and cites evidence verified against the CV, keeping the result explainable.

● RESTful API & System Integration: designing a versioned REST API to let the AI scoring microservice, web frontend, and mobile-friendly candidate portal interact independently and be deployed/scaled separately.

● Marketplace & Incentive Design: modeling a two-sided marketplace (clients posting jobs vs. affiliates/candidates supplying candidates) with commission-based incentive alignment and dispute-resolution rules (first-submission timestamp precedence).

● Software Testing & Deployment: unit/integration testing of the matching engine's scoring consistency, load testing of the job-search and duplicate-check paths, and CI/CD-based containerized deployment.

Technologies *

● Backend: ASP.NET Core Web API (or Node.js/NestJS) with Entity Framework Core, SignalR for real-time interview/notification updates

● Frontend: ~~ReactJS with Material UI~~ReactJS with TypeScript, Ant Design and Tailwind CSS for the public site, company/admin dashboards, and job-filter UI

● AI Service: Python microservice (FastAPI) ~~using Sentence-BERT / TF-IDF embeddings for CV-JD similarity scoring,~~using Sentence-BERT embeddings and an LLM (Google Gemini, configurable) for CV-JD scoring, exposed via internal REST API

● Database: PostgreSQL for transactional data (jobs, candidates, applications, commissions), with a seed table for the LinkedIn-based Industry/Company-Size taxonomy

● Storage: ~~MinIO/S3-compatible object storage for CV files and company logos/banners~~Cloudflare R2 (S3-compatible) object storage for CV files, generated PDFs and signed agreements; Resend for transactional e-mail

● Infrastructure: ~~Docker Compose for local development, CI/CD pipeline, and role-based JWT authentication~~Docker images deployed on Render (backend) and Vercel (frontend), GitHub Actions CI/CD, and role-based JWT authentication

f) Products (Expected Deliverables):

● HR Connect Web Portal (public site: home, company directory, job listings, service pages, contact form)

● Client Company Dashboard (job posting wizard, candidate shortlist review, interview tracking)

● Candidate Portal (profile/CV builder with 3 creation modes, job search & apply, job-match notifications)

● Affiliate/OPR Hub Dashboard (my jobs, candidate submission, commission ledger, performance rating)

● AI CV-JD Matching microservice with color-coded scoring API

● Admin console (company/job approval, duplicate dispute resolution, commission configuration, reporting dashboard)

● Technical documentation: ERD, API specification, and deployment guide

g) Proposed Tasks:

Work Package 1 - Requirement Analysis & System Design

● Formalize use cases from the HR Connect wireframe/functional specification for ~~all four actor roles~~all actor roles

● Design ERD and REST API specification covering Job, Candidate, Application, Affiliate, and Commission entities

Work Package 2 - Company & Job Management Module

● Implement company registration/approval and job posting workflow (Details, Instructions, Objectives, Engagement)

● ~~Implement job status lifecycle and public/affiliate visibility rules~~Implement the job status lifecycle, service agreement OTP signing and role/service-type visibility rules

Work Package 3 - Candidate & Application Pipeline

● Implement candidate profile/CV creation (3 modes) and CV submission with duplicate detection

● Implement interview scheduling, outcome logging, and ~~offer/probation tracking~~offer, start-of-work and warranty tracking

Work Package 4 - AI Matching Engine

● Build and evaluate the CV-JD semantic similarity model and ~~Must-have/Should-have keyword filter~~per-requirement LLM assessment with verified evidence

● Expose the scoring engine via internal API and integrate the color-coded ranking into the review dashboard

Work Package 5 - Affiliate & Commission System

● Implement affiliate registration, first-submission attribution, and commission rule engine

● Implement affiliate performance rating and payout ledger with reporting

Work Package 6 - Testing, Integration & Deployment

● Unit/integration testing across all modules, load testing of search/duplicate-check paths

● Containerize services with Docker, configure CI/CD, and prepare final demo dataset and deployment

4. Other comments (propose all relative things if have):

.....

.....

.....

HCM, date 19/08/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*